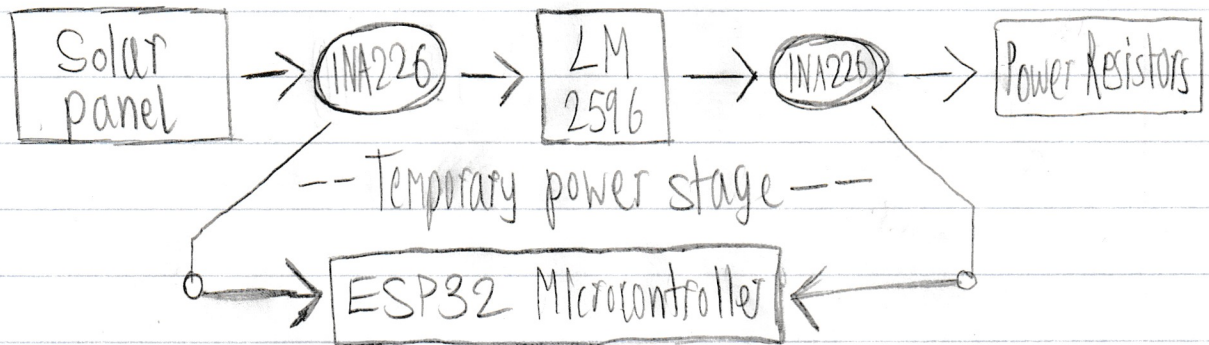


MPPT charge Controller: Prototype #1



• Solderless breadboard only

*What is Progressing?:

- Solar panel integration into this power stage already
- INA226 studying change in power provided by panel + a (somewhat) stable buck conversion output
- Studying output by using power resistor as an output load
- ESP32 data-logging, no controls quite yet

*Cons of this Design; what is to be changed?:

- Biggest: Buck converter is a commercial LM2596, which is functional but not controllable power-wise
- Only voltage control, so we cannot apply MPPT
- ESP32 isn't doing any activation or controlling due to ↑ points

*Next prototype: Build custom buck converter, ensure it works, and switch out the LM2596 with the custom buck converter